

Nguyen Khac Minh Tan

Full-Stack Developer

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SUMMARY

I build systems that handle real traffic at real scale. Production software that processes millions of transactions, serves millions of users, and stays up when things go wrong. My work sits at the intersection of distributed systems, real-time data pipelines, and platform engineering.

EXPERIENCE

Graduation thesis

2/2026 - 6/2026

Develop a project to monitor and predict traffic density using a microservice system.

Full-Stack Developer — DataFlow Inc

2018 - 2021

Built real-time analytics platforms and streaming data pipelines. Reduced data lag from 24h to under 30s using Flink and Kafka.

EDUCATION

- B.S. Software Engineer — Van Lang University (2026)

PROJECTS

Payment Gateway Migration (fintech)

2024-01-15

Problem: Legacy monolith processing 2M+ daily transactions with 0.8% failure rate and 12s avg latency.

Actions: Designed event-driven microservice architecture; Implemented saga pattern for distributed transactions; Built circuit breaker with adaptive thresholds

Results: 99.97% uptime | 180ms p99 latency | 0.01% failure rate

Tech: Go, Kafka, PostgreSQL, K8s

Real-time Analytics Pipeline (data)

2024-03-22

Problem: Batch ETL jobs causing 24h data lag, making real-time decisions impossible for ops team.

Actions: Replaced batch with streaming using Flink; Built custom windowed aggregations; Implemented exactly-once semantics

Results: <30s data lag | 500K events/sec | 40% cost reduction

Tech: Flink, Kafka, ClickHouse, Grafana

Auth Platform Redesign (security)

2024-05-10

Problem: Homegrown auth with 3K LoC, no MFA, and session fixation vulnerabilities across 12 services.

Actions: Implemented OIDC-compliant auth server; Added FIDO2/WebAuthn MFA support; Built token introspection layer for service mesh

Results: 0 auth vulns | 2M+ daily logins | <50ms token validation

Tech: Rust, Redis, OAuth2, WebAuthn

E-commerce Platform (retail)

2024-06-01

Problem: Black Friday traffic spikes causing cascading failures, \$2M lost revenue per outage minute.

Actions: Implemented adaptive rate limiting; Built inventory reservation with TTL locks; Created chaos engineering test suite

Results: 0 outages | 10x peak traffic | \$18M BF revenue

Tech: Node.js, Redis, RabbitMQ, AWS

CI/CD Pipeline Overhaul (devops)

2024-07-14

Problem: Build times averaging 45min, flaky tests at 30% failure rate, deploys taking 2h with manual gates.

Actions: Parallelized test suites with deterministic test containers; Implemented canary deployments with automated rollback; Built custom test impact analysis to skip unaffected suites

Results: 4min build time | <1% flaky rate | 15min deploy time

Tech: Go, K8s, Terraform, ArgoCD

Healthcare Data Platform (healthcare)

2024-08-20

Problem: HIPAA-compliant data silos across 8 hospitals, patient records taking 72h to sync between systems.

Actions: Built FHIR-compliant data exchange layer; Implemented field-level encryption with key rotation; Created audit trail with immutable event log

Results: <5min record sync | 100% HIPAA compliance | 8 hospitals live

Tech: Rust, PostgreSQL, Kafka, AWS

Multi-tenant SaaS Platform (saas)

2024-09-05

Problem: Single-tenant architecture hitting scaling ceiling at 50 customers, onboarding taking 3 weeks per tenant.

Actions: Designed row-level security with tenant isolation; Built self-service onboarding with infrastructure-as-code; Implemented tenant-aware connection pooling

Results: 500+ tenants | 15min onboarding | 60% infra cost saved

Tech: TypeScript, PostgreSQL, Redis, Docker

Fraud Detection Engine (fintech)

2024-10-12

Problem: Rule-based fraud detection missing 35% of fraud cases while generating 90% false positive rate.

Actions: Built real-time feature store from transaction streams; Implemented ML model serving with A/B testing framework; Created explainability layer for regulatory compliance

Results: 92% detection rate | <8% false positive | <100ms scoring latency

Tech: Python, Flink, Kafka, Redis

IoT Fleet Management (iot)

2024-11-01

Problem: 50K+ devices with unreliable connectivity, 40% telemetry data loss, and no remote update capability.

Actions: Built MQTT broker cluster with store-and-forward; Implemented delta-based OTA update system; Created device shadow for offline state sync

Results: 99.5% data delivery | 50K+ devices managed | <2GB update size

Tech: Go, MQTT, DynamoDB, AWS

Content Delivery Network (infra)

2024-11-18

Problem: Origin server handling 100% of traffic, 800ms avg TTFB for APAC users, \$500K/month bandwidth cost.

Actions: Deployed edge nodes across 12 PoPs globally; Built cache invalidation with surrogate key architecture; Implemented origin shielding with request coalescing

Results: <50ms APAC TTFB | 95% cache hit rate | \$120K monthly cost

Tech: Rust, Nginx, Terraform, AWS

Search Platform Rebuild (data)

2024-12-03

Problem: MySQL LIKE queries on 50M records, 12s search latency, zero relevance tuning capability.

Actions: Migrated to Elasticsearch with custom analyzers; Built learning-to-rank pipeline from click logs; Implemented type-ahead with prefix trie index

Results: <80ms search latency | 3x CTR improvement | 50M documents indexed

Tech: Python, Elasticsearch, Kafka, K8s

Chat & Notification System (saas)

2025-01-08

Problem: Polling-based messaging consuming 60% of API capacity, 5s message delivery latency, no offline support.

Actions: Built WebSocket gateway with connection draining; Implemented message queue with delivery guarantees; Created offline-first client with CRDT sync

Results: <200ms delivery | 80% API load reduced | 100K concurrent users

Tech: TypeScript, Redis, WebSocket, CRDTs

Compliance Automation Platform (security)

2025-02-14

Problem: SOC2 audit requiring 3 months of manual evidence collection, 200+ controls tracked in spreadsheets.

Actions: Built continuous control monitoring from infrastructure APIs; Implemented evidence auto-collection pipeline; Created drift detection against compliance baselines

Results: 2 weeks audit prep | 200+ controls automated | 98% evidence coverage

Tech: Go, Terraform, PostgreSQL, AWS

Video Transcoding Pipeline (media)

2025-03-01

Problem: Manual transcoding workflow, 8h per video, no adaptive bitrate, \$200K/year in compute waste.

Actions: Built event-driven transcoding with per-title encoding; Implemented adaptive bitrate ladder optimization; Created GPU-accelerated preview generation

Results: 15min per video | 40% storage saved | 4K max resolution

Tech: Go, FFmpeg, AWS, Kafka

CERTIFICATIONS

- Olympic PMNM Contest 2025 — BTC OLYMPIC TIN HOC SINH VIEN VIET NAM (12/12/2025)
- ICPC Vietnam Southern Provincial PC — William B.Poucher, Ph.D ICPC Executive Director (11/10/2025)
- ICPC Vietnam National PC — William B.Poucher, Ph.D ICPC Executive Director (8/11/2025)